TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	55.3 ± 2.00
Correlation Stability	70.6 ± 3.38
Distribution Consistency	57.9 ± 2.47
Entropy Preservation	22.6 ± 0.66
Mean Z Score Drift	1.2 ± 0.12
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	2.0 ± 8.91
Novelty Score	0.9 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: breast_cancer_coimbra.csv

1. Dataset Portrait

Attribute	Value
Rows	116
Columns	10
Numeric features	10
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	57.302	72.043	+14.742	25.7%	Drift
BMI	27.582	28.939	+1.357	4.9%	OK
Glucose	97.793	144.293	+46.500	47.5%	Drift
Insulin	10.012	28.010	+17.998	179.8%	Drift
HOMA	2.695	14.095	+11.400	423.0%	Drift
Leptin	26.615	41.689	+15.074	56.6%	Drift
Adiponectin	10.181	17.871	+7.690	75.5%	Drift
Resistin	14.726	34.938	+20.212	137.3%	Drift
MCP.1	534.647	1113.766	+579.119	108.3%	Drift
Classification	0.552	0.391	-0.160	29.1%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	55.3 ± 2.00
Correlation Stability (*2)	70.6 ± 3.38
Distribution Consistency (*3)	57.9 ± 2.47
Entropy Preservation (*4)	22.6 ± 0.66
Mean Z Score Drift (*5)	1.2 ± 0.12
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00

Feature Accuracy (*10)	2.0 ± 8.91
Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4625	0.5166666666666667	0.5166666666666666	0.05416666666666666	
F1	0.5361601301923884	0.28197478991596636	0.5788692045967642	0.04270907440437593	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.725	0.4916666666666667	0.6791666666666667	-0.04583333333333335	
F1	0.7359528619528619	0.28072841506665036	0.7004314526053657	-0.0355214093474963	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6416666666666667	0.4833333333333333	0.5083333333333333	-0.13333333333333333	*
F1	0.5557236965932618	0.17622926093514343	0.4064421723787095	-0.14928152421455693	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7333333333333332	0.4833333333333333	0.6041666666666666	-0.1291666666666667	*
F1	0.7439199241538071	0.23385498402526267	0.6220750264228525	-0.12184489773095471	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4958333333333333	0.5208333333333334	0.5041666666666667	0.008333333333333331	
F1	0.6547289527475286	0.3591347240650646	0.5584966952846554	-0.09623225746287319	

Model: AB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7458333333333333	0.5	0.7125	-0.03333333333333331	
F1	0.754839950419102	0.2721417719677568	0.7286181274442144	-0.02622182297488762	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.725	0.4833333333333334	0.6833333333333333	-0.04166666666666667	
F1	0.7511736533167108	0.21095116458494068	0.7060212357164527	-0.04515241760025821	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7583333333333334	0.4875	0.6916666666666667	-0.0666666666666667	
F1	0.7810334757834757	0.2645387996047225	0.7235773406032027	-0.05745613518027311	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7083333333333334	0.4875	0.6	-0.10833333333333335	
F1	0.7153156425450278	0.30978441287702296	0.5505561927004052	-0.1647594498446226	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4791666666666667	0.5458333333333333	0.5791666666666667	0.1	
F1	0.606269362684906	0.5611603015696452	0.5611517361219878	-0.04511762656291827 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	48.0	23.5	70.0	2.707	0.467408667	8.8071	9.7024	7.99585	417.114	0.0
2	83.0	20.69049454	92.0	3.115	0.706897333	8.8438	5.429285	4.06405	468.786	0.0
3	82.0	23.12467037	91.0	4.498	1.009651067	17.9393	22.43204	9.27715	554.697	0.0
4	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0
5	86.0	21.11111111	92.0	3.549	0.8053864	6.6994	4.81924	10.57635	773.92	0.0
6	49.0	22.85445769	92.0	3.226	0.732086933	6.8317	13.67975	10.3176	530.41	0.0
7	89.0	22.7	77.0	4.69	0.890787333	6.964	5.589865	12.9361	1256.083	0.0
8	76.0	23.8	118.0	6.47	1.883201333	4.311	13.25132	5.1042	280.694	0.0
9	73.0	22.0	97.0	3.35	0.801543333	4.47	10.358725	6.28445	136.855	0.0
10	75.0	23.0	83.0	4.952	1.013839467	17.127	11.57899	7.0913	318.302	0.0
11	34.0	21.47	78.0	3.469	0.6674356	14.57	13.11	6.92	354.6	0.0
12	29.0	23.01	82.0	5.663	1.145436133	35.59	26.72	4.58	174.8	0.0
13	25.0	22.86	82.0	4.09	0.827270667	20.45	23.67	5.14	313.73	0.0
14	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
15	38.0	23.34	75.0	5.782	1.06967	15.26	17.95	9.35	165.02	0.0
16	44.0	20.76	86.0	7.553	1.6	14.09	20.32	7.64	63.61	0.0
17	47.0	22.03	84.0	2.869	0.59	26.65	38.04	3.32	191.72	0.0
18	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
19	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
20	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	24.0	18.37	60.0	58.46	25.05034187	90.28	1.65602	3.21	45.843	0.0
2	24.0	38.57875854	60.0	58.46	25.05034187	90.28	1.65602	3.21	45.843	0.0
3	89.0	38.57875854	201.0	2.432	25.05034187	90.28	1.65602	82.1	1698.44	1.0
4	89.0	38.57875854	201.0	58.46	25.05034187	4.311	38.04	3.21	45.843	1.0
5	89.0	18.37	201.0	2.432	0.467408667	4.311	38.04	82.1	1698.44	0.0
6	89.0	18.37	60.0	58.46	25.05034187	4.311	1.65602	3.21	1698.44	0.0
7	64.0	22.22222222	98.0	5.7	1.37788	12.1905	4.783985	13.91245	395.976	1.0
8	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
9	68.0	21.08281329	102.0	6.2	1.55992	9.6994	8.574655	13.74244	448.799	1.0
10	42.0	21.35991456	93.0	2.999	0.6879706	19.0826	8.462915	17.37615	321.919	1.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	86.0	37.03560819	70.0	2.432	25.05034187	89.27	1.65602	3.21	1698.44	1.0
2	24.0	37.03560819	70.0	2.432	25.05034187	89.27	1.65602	3.21	1698.44	1.0
3	86.0	18.37	199.0	2.432	0.467408667	89.27	1.65602	3.21	1698.44	1.0
4	86.0	18.37	70.0	2.432	0.467408667	89.27	38.04	3.21	1698.44	0.0
5	24.0	18.37	199.0	58.46	25.05034187	4.47	38.04	82.1	1698.44	1.0
6	86.0	18.37	70.0	2.432	0.467408667	89.27	1.65602	3.21	1698.44	0.0
7	50.0	38.57875854	106.0	6.703	1.752611067	46.6401	4.667645	11.78388	887.16	0.0
8	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
9	38.0	22.4996371	95.0	5.261	1.232827667	8.438	4.77192	15.73606	199.055	1.0
10	81.0	31.64036818	100.0	9.669	2.38502	38.8066	10.636525	29.5583	426.175	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	18.37	70.0	2.74	25.05034187	90.28	1.65602	3.21	45.843	0.0
2	25.0	18.37	70.0	2.74	25.05034187	90.28	1.65602	3.21	45.843	1.0
3	89.0	37.109375	70.0	58.46	0.570392	90.28	38.04	82.1	45.843	1.0
4	25.0	18.37	70.0	2.74	25.05034187	4.311	1.65602	3.21	45.843	1.0
5	25.0	18.37	201.0	58.46	0.570392	4.311	38.04	82.1	45.843	1.0
6	25.0	37.109375	70.0	2.74	25.05034187	90.28	1.65602	3.21	45.843	1.0
7	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
8	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
9	71.0	25.51020408	112.0	10.395	2.871792	19.0653	5.4861	42.7447	799.898	1.0
10	49.0	21.36752137	78.0	2.64	0.507936	6.3339	3.886145	22.94254	737.672	1.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	38.57875854	201.0	58.46	25.05034187	90.28	1.65602	3.21	45.843	0.0
2	89.0	38.57875854	201.0	58.46	0.467408667	90.28	1.65602	3.21	45.843	1.0
3	89.0	38.57875854	201.0	58.46	0.467408667	4.311	1.65602	82.1	45.843	1.0
4	25.0	18.37	60.0	2.432	25.05034187	90.28	38.04	3.21	45.843	0.0
5	89.0	38.57875854	201.0	58.46	0.467408667	4.311	38.04	82.1	1698.44	1.0
6	89.0	38.57875854	201.0	58.46	0.467408667	90.28	1.65602	82.1	45.843	1.0
7	75.0	30.48	152.0	7.01	2.628282667	50.53	10.06	11.73	99.45	1.0
8	60.0	26.34929208	103.0	5.138	1.305394533	24.2998	2.19428	20.2535	378.996	0.0
9	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
10	72.0	23.62	105.0	4.42	1.14478	21.78	17.86	4.82	195.94	1.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	24.0	18.37	60.0	58.46	25.05034187	4.311	2.19428	3.21	45.843	0.0
2	24.0	18.37	60.0	58.46	25.05034187	4.311	2.19428	55.2153	1698.44	0.0
3	89.0	38.57875854	60.0	2.432	25.05034187	4.311	2.19428	55.2153	1698.44	0.0
4	89.0	18.37	60.0	58.46	25.05034187	4.311	38.04	3.21	45.843	0.0
5	24.0	18.37	60.0	2.432	25.05034187	4.311	2.19428	55.2153	1698.44	0.0
6	89.0	38.57875854	60.0	2.432	0.507936	4.311	2.19428	55.2153	1698.44	0.0
7	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
8	48.0	28.125	90.0	2.54	0.56388	15.5325	10.22231	16.11032	1698.44	1.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	18.37	60.0	58.46	25.05034187	90.28	38.04	3.21	1698.44	0.0
2	24.0	18.37	60.0	58.46	25.05034187	4.311	1.65602	3.21	45.843	0.0
3	24.0	18.37	60.0	2.432	25.05034187	4.311	1.65602	82.1	45.843	1.0
4	24.0	18.37	60.0	58.46	25.05034187	4.311	38.04	3.21	45.843	0.0
5	24.0	18.37	199.0	58.46	25.05034187	4.311	1.65602	3.21	45.843	0.0
6	24.0	38.57875854	199.0	58.46	25.05034187	4.311	1.65602	82.1	45.843	1.0
7	74.0	28.65013774	88.0	3.012	0.6538048	31.1233	7.65222	18.35574	572.401	1.0
8	65.0	32.05	97.0	5.73	1.370998	61.48	22.54	10.33	314.05	1.0
9	76.0	27.2	94.0	14.07	3.262364	35.891	9.34663	8.4156	377.227	0.0
10	65.0	29.666548	85.0	14.649	3.071407	26.5166	7.28287	19.46324	1698.44	1.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	24.0	38.57875854	60.0	2.432	25.05034187	90.28	38.04	82.1	1698.44	1.0
2	24.0	18.37	60.0	58.46	25.05034187	4.311	1.65602	82.1	1698.44	1.0
3	24.0	18.37	60.0	58.46	25.05034187	4.311	1.65602	82.1	45.843	1.0
4	24.0	18.37	60.0	2.432	0.507936	4.311	1.65602	82.1	45.843	0.0
5	24.0	18.37	60.0	58.46	25.05034187	90.28	1.65602	82.1	1698.44	1.0
6	24.0	18.37	60.0	58.46	25.05034187	4.311	1.65602	82.1	45.843	0.0
7	35.0	30.27681661	84.0	4.376	0.9067072	39.2134	9.048185	16.43706	733.797	0.0
8	69.0	32.5	93.0	5.43	1.245642	15.145	11.78796	11.78796	270.142	0.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	49.0	29.77777778	70.0	8.396	1.449709333	51.3387	10.73174	20.76801	602.486	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	18.37	60.0	58.46	0.467408667	4.311	1.65602	3.21	45.843	0.0
2	24.0	38.57875854	201.0	58.46	20.6307338	83.4821	1.65602	3.21	1698.44	1.0
3	24.0	38.57875854	201.0	58.46	20.6307338	83.4821	1.65602	82.1	1698.44	0.0
4	24.0	38.57875854	201.0	58.46	20.6307338	83.4821	1.65602	82.1	1698.44	1.0
5	89.0	38.57875854	201.0	58.46	20.6307338	4.311	1.65602	82.1	1698.44	0.0
6	89.0	18.37	60.0	2.432	0.467408667	4.311	1.65602	3.21	45.843	0.0
7	45.0	37.03560819	83.0	6.76	1.383997333	39.9802	4.617125	8.70448	586.173	0.0
8	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
9	46.0	22.21	86.0	36.94	7.836205333	10.16	9.76	5.68	312.0	1.0
10	85.0	27.68877813	196.0	51.814	25.05034187	70.8824	7.901685	55.2153	1078.359	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	38.57875854	70.0	58.46	20.6307338	90.28	1.65602	82.1	1698.44	0.0
2	24.0	18.37	70.0	58.46	20.6307338	4.311	38.04	3.21	90.09	0.0
3	24.0	18.37	70.0	58.46	20.6307338	4.311	38.04	3.21	1698.44	1.0
4	24.0	18.37	70.0	2.432	0.467408667	4.311	38.04	3.21	90.09	0.0
5	89.0	38.57875854	201.0	2.432	0.467408667	90.28	38.04	3.21	1698.44	0.0
6	89.0	38.57875854	201.0	58.46	20.6307338	90.28	1.65602	82.1	1698.44	0.0
7	78.0	25.3	60.0	3.508	0.519184	6.633	10.567295	4.6638	209.749	0.0
8	52.0	30.8012487	87.0	30.212	6.4834952	29.2739	6.26854	24.24591	764.667	1.0
9	45.0	20.26	92.0	3.44	0.780650667	7.65	16.67	7.84	193.87	1.0
10	59.0	28.67262608	77.0	3.188	0.605507467	17.022	16.44048	31.6904	910.489	1.0

TabuLLM: AI Benchmark Report

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	18.67	60.0	2.432	0.467408667	90.28	38.04	82.1	45.843	0.0
2	24.0	38.57875854	201.0	58.46	25.05034187	4.311	1.65602	3.27	1698.44	0.0
3	89.0	38.57875854	201.0	58.46	0.467408667	90.28	1.65602	82.1	45.843	0.0
4	24.0	38.57875854	201.0	2.432	25.05034187	4.311	38.04	3.27	1698.44	1.0
5	89.0	18.67	201.0	2.432	0.467408667	90.28	1.65602	82.1	45.843	1.0
6	89.0	18.67	201.0	2.432	0.467408667	90.28	38.04	82.1	45.843	1.0
7	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
8	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0
9	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
10	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0